

# Donisha Smith

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## EDUCATION

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### Florida International University

Aug 2025

PhD in Cognitive Neuroscience

Dissertation: *"Neurobiological Mechanisms of Physics Learning"*

Advisor: Dr. Angela Laird

### Florida International University

Aug 2022

M.Sc. in Cognitive Neuroscience

Thesis: *"Task-based attentional and default mode connectivity associated with science and math anxiety profiles among university physics students"*

Advisor: Dr. Angela Laird

### Florida International University

Apr 2018

B.Sc. in Biological Sciences (Cum Laude), Minor in Chemistry, Quantifying Biology in the Classroom (QBIC) track, Honor's College

## EXPERIENCE

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### Postdoctoral Researcher, Johns Hopkins University School of Medicine

Oct 2025-Current

- Developed an automated pipeline to convert multi-session pharmacological NIfTI datasets without source DICOMS and JSON sidecars into BIDS format by reconstructing missing MRI acquisition metadata from vendor documentation and NIfTI headers
- Designed and deployed a multi-step fMRI analysis pipeline on HPC, orchestrating SLURM job dependencies for GLM, gPPI, parametric/nonparametric group inference, and automated MNI region labeling using AFNI and FSL
- Established the lab's first scalable neuroimaging infrastructure by migrating local GUI-based workflows to reproducible, containerized HPC pipelines using Apptainer with version-pinned tools (AFNI, FSL, fMRIPrep)

### Graduate Researcher, Florida International University

Aug 2019-Aug 2025

- Developed a k-means clustering pipeline to characterize dynamic brain state changes associated with learning in longitudinal fMRI timeseries data
- Built post-processing pipelines using fMRIPrep and MRIQC to extract and denoise timeseries data, minimizing motion artifacts and increasing signal-to-noise for downstream statistical modeling
- Created statistical models (e.g. Linear Mixed Effects, Structural Equation Modeling, Latent Profile Analysis) to examine brain-behavioral relationships

- Analyzed staff demographic data for the ABCD Study consortium using R, implementing hypothesis testing and data visualization to inform decisions for a multi-site research task force
- Designed and delivered an introductory workshop on R programming and Tidyverse for the Diversity, Equity, and Inclusion program, instructing 10+ graduate students on data manipulation and visualization
- Provided statistical consultation on collaborative projects, applying latent and linear methods to neuroimaging and tabular data

#### **Graduate Teaching Assistant, Florida International University**

**Jan 2020-May 2022**

- Provided instructional support and targeted tutoring for several undergraduate psychology statistics courses

#### **Medical Laboratory Scientist I, Florida Department of Health**

**Jan 2019-Jul 2019**

- Conducted confirmatory diagnostic testing for HIV and syphilis for public health surveillance programs

### **SOFTWARE**

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#### **NIFTI2BIDS** | [Github](#) | [Docs](#)

**Oct 2025-Current**

- Developed a Python package for post-hoc BIDS conversion of NIfTI datasets lacking source DICOMs, reconstructing JSON sidecars with acquisition metadata derived from image headers and vendor-specific documentation
- Built configurable parsers for Presentation and E-Prime behavioral logs, automating timing and accuracy extraction for block and event-related fMRI designs into BIDS-compliant events files
- Implemented automated cross-platform testing across Python 3.10-3.14 on Linux, macOS, and Windows, maintaining >90% code coverage

#### **NeuroCAPs** | [Github](#) | [Docs](#) | [Demo](#) | [JOSS Paper](#) | [JHU OSPO Catalog](#)

**Jan 2024-Current**

- Developed a Python package for fMRI brain state identification using k-means, achieving 5x faster multi-subject processing via multiprocessing with checkpointing support
- Incorporated clustering validation (e.g. Davies-Bouldin, Silhouette), KD-tree interpolation for volumetric-to-surface mapping, temporal dynamics metrics (e.g. transition probabilities), and an interactive visualization suite
- Deployed via Docker with headless VTK rendering and CLI/Jupyter interfaces; maintained >90% test coverage across OS platforms using pytest and GitHub Actions

#### **vswift** | [Github](#)

**Apr 2023-Current**

- Created an R package for ML model evaluation, implementing custom stratified sampling, cross-validation, and nested cross-validation for hyperparameter tuning, with parallel processing support at the fold-level

- Designed a unified interface supporting classification algorithms (e.g., Regularized Logistic Regression, SVM, XGBoost, Neural Networks) while preserving algorithm-specific parameter tuning
- Incorporated automated missing data imputation and multi-metric performance assessment (precision, recall, F1) with CLI output and visualizations, including ROC and Precision-Recall curves with AUC scores using trapezoidal rule

## WORKSHOPS

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### High Performance Computing Workshop: Machine Learning and Big Data

Jan 2025

- Attended an interactive workshop focusing on applying big data analytics with Spark and implementing deep learning using TensorFlow.

## PUBLICATIONS

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- [1] **Smith, D. D.**, Bartley, J. E., Peraza, J. A., Bottenhorn, K. L., Nomi, J. S., Uddin, L. Q., Riedel, M. C., Salo, T., Laird, R. W., Pruden, S. M., Sutherland, M. T., Brewe, E., & Laird, A. R. (2025). Dynamic reconfiguration of brain coactivation states associated with active and lecture-based learning of university physics. *Npj Science of Learning*, 10(1), 55. <https://doi.org/10.1038/s41539-025-00348-9>
- [2] **Smith, D.** (2025). NeuroCAPs: A Python Package for Performing Co-Activation Patterns Analyses on Resting-State and Task-Based fMRI Data. *Journal of Open Source Software*, 10(112), 8196. <https://doi.org/10.21105/joss.08196>
- [3] Pintos Lobo, R., Peraza, J. A., Salo, T., Meca, A., **Smith, D. D.**, Feeney, K. E., Schmardeer, K. M., Sutherland, M. T., Gonzalez, R., Musser, E. D., & Laird, A. R. (2025). Social profiles among youth with attention-deficit/hyperactivity disorder (ADHD): Evidence from the ABCD study. *Developmental Cognitive Neuroscience*, 75, 101591. <https://doi.org/10.1016/j.dcn.2025.101591>
- [4] **Smith D. D.**, Meca A, Bartley JE, Riedel MC, Salo T, Peraza JA, Bottenhorn KL, Laird RW, Pruden SM, Sutherland MT, Brewe E, Laird AR (2023). Task-based attentional and default mode connectivity associated with science and math anxiety profiles among university physics students. *Trends in Neuroscience and Education*. <https://doi.org/10.1016/j.tine.2023.100204>
- [5] Lobo, R. P., Bottenhorn, K. L., Riedel, M. C., Toma, A. I., Hare, M. M., **Smith, D. D.**, Moor, A. C., Cowan, I. K., Valdes, J. A., Bartley, J. E., Salo, T., Boevig, E. R., Pankey, B., Sutherland, M. T., Musser, E. D., & Laird, A. R. (2022). Neural systems underlying RDoC social constructs: An activation likelihood estimation meta-analysis. *Neuroscience & Biobehavioral Reviews*. <https://doi.org/10.1016/j.neubiorev.2022.104971>

## PREPRINTS

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- [1] **Smith D. D.**, Bartley JE, Riedel MC, Salo T, Peraza JA, Bottenhorn KL, Laird RW, Pruden SM, Sutherland MT, Brewe E, Laird AR (2025). Hippocampal functional connectivity changes associated with active and lecture-based physics learning. bioRxiv.  
<https://doi.org/10.1101/2025.09.22.677908>
- [2] Hampson, C. L., Peraza, J. A., Guerrero, L. M., Bottenhorn, K. L., Riedel, M. C., Almuquin, F., **Smith, D. D.**, Schmarader, K. M., Crooks, K. E., Lobo, R. P., Sutherland, M. T., Musser, E. D., Dai, Y., Agarwal, R., Saeed, F., & Laird, A. R. (2025). Habenula alterations in resting state functional connectivity among autistic individuals.  
<https://doi.org/10.1101/2025.05.14.653992>

## PRESENTATIONS

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- [1] Hampson, C.L., Peraza, J.P., Guerrero, L.M., Bottenhorn, K.L., Riedel, M.C., Almuquin, F., **Smith, D.D.**, Schmarader, K.M., Musser, E.D., Dai, Y., Agarwal, R., Saeed, F., Sutherland, M.T., Laird, A.R., Habenula alterations in resting state functional connectivity in autism. (Accepted). Poster at the 31th Annual Meeting of the Organization for Human Brain Mapping, Brisbane, Australia.
- [2] Hampson, C. L., Peraza, J. A., Guerrero, L. M., Bottenhorn, K. L., Riedel, M. C., Almuqhim, F., Saeed, F., **Smith, D. D.**, Schmarader, K. M., Crooks, K. E., Viera Perez, P. M., Musser, E. D., Dai, Y., Agarwal, R., Sutherland, M. T., & Laird, A. R. (2024). Habenula alterations in resting state functional connectivity in autism spectrum disorder. Poster presented at the University of Miami's 34th Annual Neuroscience Research Day, Miami, FL.
- [3] **Smith D. D.**, Bartley JE, Peraza JA, Riedel MC, Salo T, Bottenhorn KL, Laird RW, Pruden SM, Sutherland MT, Brewe E, Laird A.R.. Longitudinal Changes in Dynamic Functional Connectivity Associated with Physics Learning. Presented on May 3, 2024, at the Florida Consortium on the Neurobiology of Cognition; Miami, Florida.
- [4] **Smith D. D.**, Bartley JE, Peraza JA, Riedel MC, Salo T, Bottenhorn KL, Laird RW, Pruden SM, Sutherland MT, Brewe E, Laird A.R.. Longitudinal Changes in Dynamic Functional Connectivity Associated with Physics Learning. Accepted at the 2024 Graduate Student Appreciation Week at Florida International University; Miami, Florida.
- [5] **Smith D. D.**, Meca A, Bartley JE, Riedel MC, Salo T, Peraza JA, Bottenhorn KL, Laird RW, Pruden SM, Sutherland MT, Brewe E, Laird A.R.. Task-based attention & default mode connectivity linked to STEM anxiety in university students. Presented on July 23, 2023, at the 29th annual meeting of the Organization for Human Brain Mapping; Montréal, Canada.

## PROFESSIONAL ASSOCIATIONS

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Organization for Human Brain Mapping (OHBM)

Jun 2023-Current

## REVIEWER

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Journal of Open Source Software

Aug 2025-Current

**FELLOWSHIPS, HONORS, SCHOLARSHIPS**

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|------------------------------------------------------------------------|--------------------------|
| Dissertation Year Fellowship (\$20,000 across two semesters)           | <b>Dec 2024-Jun 2025</b> |
| DEI Doctoral Fellowship (\$3,000 over three semesters)                 | <b>Aug 2022-Aug 2023</b> |
| Recipient of Undergraduate NIGMS RISE Fellowship                       | <b>Jan 2017</b>          |
| FIU Ambassador Scholarship (Full-Tuition Merit Scholarship)            | <b>Aug 2014-May 2018</b> |
| QBIC Scholarship (\$1,000 per semester)                                | <b>Aug 2014-May 2018</b> |
| Florida Academic Scholars Award (Highest State Merit Scholarship Tier) | <b>Aug 2014-May 2018</b> |
| Volunteer Recognition Award, Memorial Hospital West                    | <b>May 2014</b>          |

**TECHNICAL SKILLS**

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**Programming Languages:** Python, R, Shell Scripting (Bash), SQL

**Computing & Development:** High-Performance Computing, CI/CD (Git, GitHub Actions, Docker, Singularity/Apptainer)

**Neuroimaging Tools:** Nilearn, fMRIPrep, AFNI, FSL, EPRIME, Presentation, PsychoPy